A FINITELY PRESENTED MODULE WHOSE STALK AT A POINT IS FREE OF RANK R IS FREE OF THAT RANK ON A NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The lifting of finitely many elements of a stalk to sections over a neighbourhood, and the spreading of surjectivity on a stalk to a neighbourhood. Used inside the counted proof (source lines 1763-1768), the second by a pointer to a lemma of this chapter. Produces a map from a finite free module onto the given module over a neighbourhood.
- The finiteness of the kernel of a surjection from a finite free module onto a finitely presented module, and the criterion for a module of finite type whose stalk at a point vanishes to vanish on a neighbourhood. Both are invoked inside the counted proof (source lines 1769-1772) as pointers to lemmas of this chapter. Shows that the kernel of the constructed map vanishes after shrinking, whence the map is an isomorphism.

Source reference. Original lemma lemma-finite-type-surjective-on-stalk: sources/stacks-modules/pinned-original-source.tex, source lines 1125–1125. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-type-surjective-on-stalk, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-kernel-surjection-finite-free-onto-finite-presentation: sources/stacks-modules/pinned-original-source.tex, source lines 1769–1769. Navigation pointer only: the counted text refers to the Stacks item lemma-kernel-surjection-finite-free-onto-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Source reference. Original lemma lemma-finite-type-stalk-zero: sources/stacks-modules/pinned-original-source.tex, source lines 1140–1140. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-type-stalk-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $x \in X$ such that $\mathcal{F}_x \cong \mathcal{O}_{X,x}^{\oplus r}$. Then there exists an open neighbourhood U of x such that $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r}$.*

Proof. Choose $s_1, \ldots, s_r \in \mathcal{F}_x$ mapping to a basis of $\mathcal{O}_{X,x}^{\oplus r}$ by the isomorphism. Choose an open neighbourhood U of x such that s_i lifts to $s_i \in \mathcal{F}(U)$. After shrinking U we see that the induced map $\psi : \mathcal{O}_U^{\oplus r} \to \mathcal{F}|_U$ is surjective (Lemma lemma-finite-type-surjective-on-stalk). By Lemma lemma-kernel-surjection-finite-free-onto-finite-presentation we see that $\mathrm{Ker}(\psi)$ is of finite type. Then $\mathrm{Ker}(\psi)_x = 0$ implies that $\mathrm{Ker}(\psi)$ becomes zero after shrinking U once more (Lemma lemma-finite-type-stalk-zero). $\square$